Reading and Making Graphs KEY

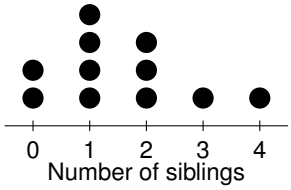
Name _____

Frequency means how many times something happens. These three tools all show frequency.

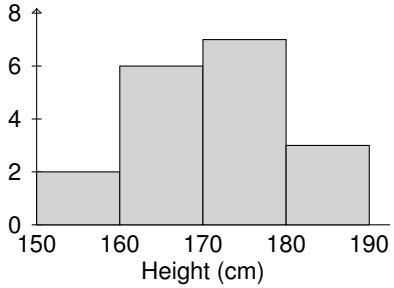
Frequency table

Siblings	Tally	Frequency
0	II	2
1	IIII	4
2	III	3
3	I	1
4	I	1

Dot plot



Histogram



Each **value** gets a row.	Each **value** gets a spot on a number line.	Values are put in **groups** (bins).
Make a tally mark for each data point, then count.	**One dot = one data point.** Stack the dots.	Bars **touch**. No gaps between groups.
Frequency column = how many.	Height of the stack = how many.	Bar height = how many in that group.

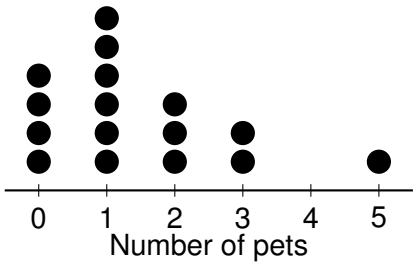
Histogram rule: If a number is **on a line** between two bars, it goes in the bar on the **right**. Example: 160 goes in the 160 to 170 bar.

The frequency table and dot plot above show the **same data**. Check: 2 + 4 + 3 + 1 + 1 = 11 students.

Every time you read a graph: read the title, read both axis labels, then check the scale.

Work with a partner. Compare answers when you finish each problem.

1. This dot plot shows how many pets some students have.



(a) How many students have exactly 2 pets? 3

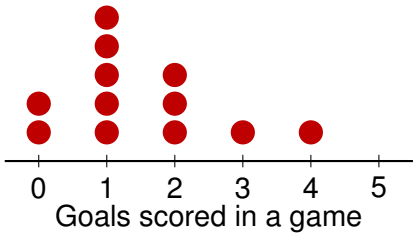
(b) How many students are in the data? 4 + 6 + 3 + 2 + 0 + 1 = 16

(c) How many students have more than 2 pets? 2 + 0 + 1 = 3

(d) What number of pets is most common? 1 pet

2. A soccer team scored these goals in 12 games. Make a dot plot.

2, 0, 1, 3, 1, 1, 4, 2, 0, 1, 2, 1



3. 20 students said how many siblings they have. Make a frequency table.

1, 2, 0, 3, 1, 1, 2, 4, 0, 2, 1, 3, 2, 1, 0, 2, 1, 5, 2, 1

Siblings	Tally	Frequency
0	III	3
1	IIIIIII	7
2	IIIIII	6
3	II	2
4	I	1
5	I	1

(a) How many students have 2 or more siblings? 6 + 2 + 1 + 1 = 10

(b) How can you use your table to check that you counted everyone? Add the frequencies. Total = 20.

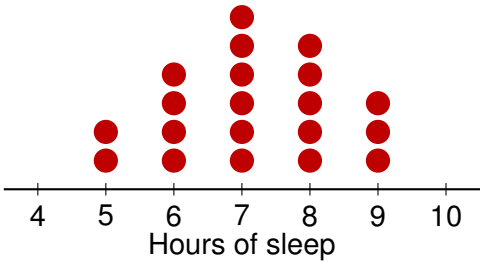
4. This frequency table shows how many hours some students slept last night.

Hours of sleep	Frequency
5	2
6	4
7	6
8	5
9	3

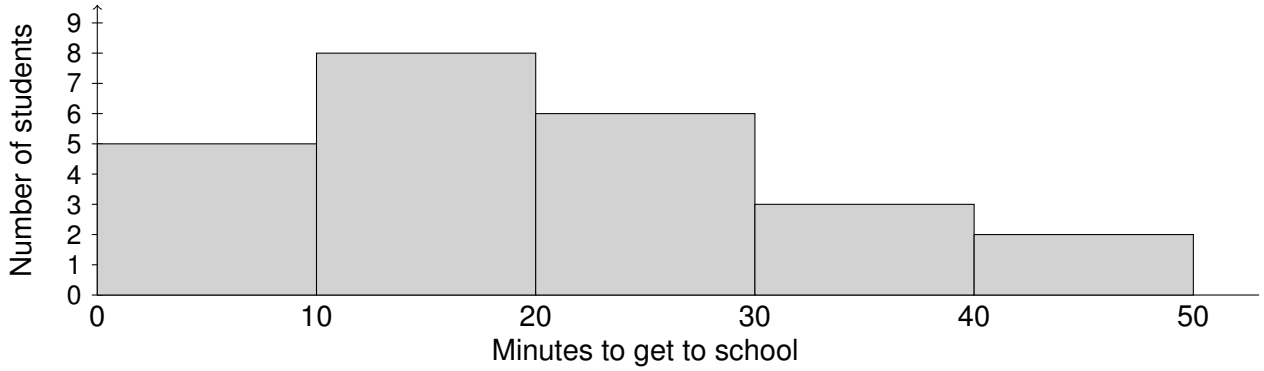
(a) How many students are in the data? 20

(b) How many students slept less than 7 hours? 2 + 4 = 6

(c) Make a dot plot of the same data.



5. This histogram shows how many minutes some students take to get to school.



(a) How many students take from 20 to 30 minutes? 6

(b) How many students take less than 20 minutes? 5 + 8 = 13

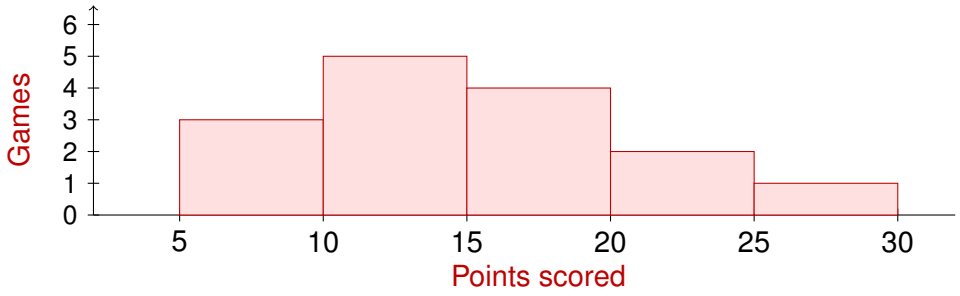
(c) How many students are in the data? 5 + 8 + 6 + 3 + 2 = 24

(d) Can you tell from this graph how many students take exactly 15 minutes? Explain.
No. The graph only shows that 8 students take from 10 to 20 minutes. It hides the exact values.

6. A basketball player scored these points in 15 games.

12, 8, 15, 21, 9, 14, 18, 10, 25, 13, 7, 16, 20, 11, 19

(a) Make a frequency table. Count how many games are in each group. Remember the histogram rule.

Points	Frequency
5 to 10	3 (7, 8, 9)
10 to 15	5 (10, 11, 12, 13, 14)
15 to 20	4 (15, 16, 18, 19)
20 to 25	2 (20, 21)
25 to 30	1 (25)

(b) Make a histogram. Label both axes.